

BIT1123 Object Oriented Programming

Assignment 1 – Self-Reflective Report

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GitHub Repository: <https://github.com/Anas777771/Object-oriented-programming-AS-1>

1. Overview

This report describes my experience completing the BIT1123 Object Oriented Programming tutorials. The activities gave me the opportunity to move from simple Java programs to more structured applications. Across Week 1 to Week 10, I practiced writing, testing, and improving Java code while working with Visual Studio Code. The exercises also helped me see how Object-Oriented Programming ideas can be applied in practical situations.

2. Week 1 – Java Fundamentals

At the beginning, I focused on the basic process of working with Java. I learned how to create source files, compile programs, and execute them from the development environment. Although these tasks were simple, they helped me understand the normal workflow required before moving to more advanced programming concepts.

3. Week 2 – Working with Classes and Objects

The next activity introduced classes and objects through a Student example. I worked with information such as the student's name, age, and GPA and created methods for actions such as displaying information, studying, and taking an examination. This exercise made the relationship between a class as a blueprint and an object as an instance much clearer to me.

4. Weeks 3–4 – Inheritance and Overriding

I then worked with a group of related classes: Person, Student, and Lecturer. Student and Lecturer were derived from Person, which demonstrated how an existing class can provide common features for other classes. I also used method overriding so that the child classes could produce their own introduction behaviour. This activity showed me how inheritance can make programs more organized and reusable.

5. Week 5 – Encapsulation

For Week 5, I developed a Student Information System using private attributes together with getter and setter methods. The information included a student ID, name, CGPA, and programme. Through

this task, I understood that data should not always be directly accessible and that methods can be used to control how information is read or changed.

6. Week 6 – Employee and Lecturer Classes

This tutorial extended my understanding of inheritance by creating an Employee class and then using Lecturer as a specialized class. Employee contained common details such as an ID and name, while Lecturer included additional information about a subject and department. The example demonstrated how a child class can build on an existing class without rewriting all of its common code.

7. Week 7 – Abstract Classes

I used an abstract Appliances class to create a simple smart-home appliance example. The program included common operations such as showing the brand, switching an appliance on or off, and performing its operation. I tested the design with a washing machine and a microwave. This helped me understand how abstraction can define shared behaviour while allowing individual subclasses to provide their own implementation.

8. Weeks 8–9 – Collections and File Processing

During Weeks 8 and 9, I created a To-Do List application using ArrayList. I entered three tasks, displayed them, stored them in a text file, and later loaded the saved information again. This was useful because it connected a Java collection with basic file input and output, showing me how a program can keep information beyond a single execution.

9. Week 10 – Java GUI Application

The final tutorial involved creating a Programming Quiz Battle with Java Swing. The interface presented a question and two possible choices. The program responded differently depending on the selected answer, showing a success message for a correct choice and a retry message for an incorrect one. Testing both outcomes helped me understand how GUI components can interact with program logic.

10. Problems I Encountered and How I Solved Them

Several technical issues appeared while completing the tutorials. These included using the wrong terminal directory, Java file and class-name mismatches, compilation errors, and Git configuration problems. I usually solved these issues by reading the compiler or terminal message carefully, checking the current folder, confirming that class and file names matched, and compiling the necessary files together. When Git required author information, I configured the username and email before committing the work. These problems improved my ability to troubleshoot instead of relying only on trial and error.

11. Development of My Java Skills

Repeated practice made me more confident with Java. I became more familiar with constructors, methods, objects, inheritance, encapsulation, abstract classes, ArrayList, file operations, and GUI components. I also became more comfortable using the terminal to compile and run programs. In addition, working with Git and GitHub gave me experience organizing and maintaining programming work.

12. What I Learned About OOP

The tutorials helped me understand the major principles of Object-Oriented Programming through actual code. Encapsulation is useful for controlling access to data, inheritance supports code reuse, overriding allows subclasses to change inherited behaviour, and abstraction helps define a common structure for related classes. Learning these ideas through separate practical tasks made them easier for me to understand than studying definitions alone.

13. Areas I Want to Improve

My next goal is to use these concepts in bigger and more realistic Java applications. I would like to become faster at finding and fixing errors and improve my use of GitHub and databases. I also want to strengthen my overall software development skills so that I can combine OOP concepts into complete projects rather than only small tutorial exercises.

14. Conclusion

Completing the BIT1123 tutorials gave me practical experience with Java and strengthened my understanding of Object-Oriented Programming. Starting with basic Java and progressing through classes, inheritance, encapsulation, abstraction, collections, file handling, and GUI programming showed me how different concepts work together. The debugging and testing involved throughout the assignment also increased my confidence in writing and improving Java programs.

15. GitHub Repository

The completed tutorial work is organized in the following GitHub repository:

<https://github.com/Anas777771/Object-oriented-programming-AS-1>